

Loan 4396-PAK: Developing Resilient Environments and Advancing Municipal Services in Punjab Project

Integrated Solid Waste Management System for Bahawalpur City



Introduction

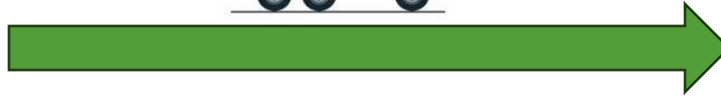
Project Title	Integrated Solid Waste Management (ISWM) System for Bahawalpur City
Location	Bahawalpur, Punjab
Sponsoring Agency	Local Government & Community Development Department
Executing Agency	Project Coordination Unit at Provincial level and City Implementation Team under Bahawalpur Solid Waste Management Company
Project Cost <i>(As per Approved PC-I)</i>	PKR 11,562.74 Million (Million USD 40.35)
Gestation Period	2024-2028 (5 Years)

Project Design Criteria

- **End-to-end solid waste solutions**
- **100% coverage of targeted population**
- **Commitment- Waste should not touch the ground.**
- **Confining waste at point of waste generation in appropriate size containers.**
- **Maximum Recovery of Recyclables.**
- **Waste treatment of maximum waste streams.**
- **Reduce green-house-gas (GHG) emissions into environment.**
- **Major shift from linear economy to circular economy.**
- **Low carbon solutions for getting Carbon credits**
- **Community behavior change.**
- **Revenue Generation and Self- Sustainability of the Project.**

Existing Scenario

Mixed Waste Collection



Open Dumping



Proposed Interventions

At-source Segregation



Material Recovery Facility



Engineered Landfill



Awareness program



Proposed Project Interventions

❑ **Waste Collection and Transportation System**

- ✓ Primary Waste Collection (Door to Door)
- ✓ Secondary Waste Collection
- ✓ Waste Transportation in Environmental friendly way

❑ **Material Recovery Facility (MRF) – Selling of Recovered Recyclables**

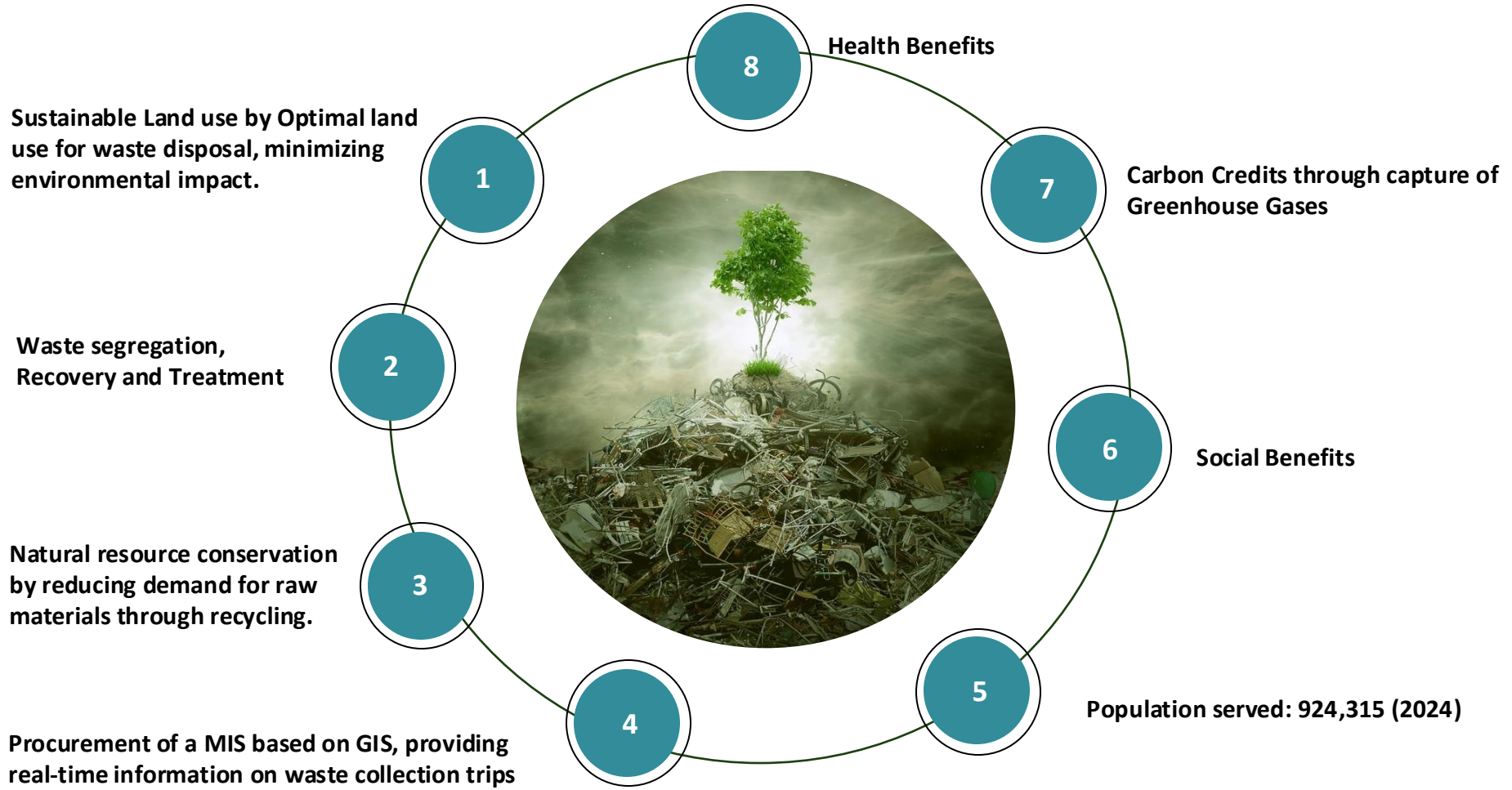
❑ **Pilot Waste Treatment**

- | | | |
|--------------------------------------|---|----------------------------------|
| ✓ Plastic waste | ➡ | Plastic Waste Recycling Facility |
| ✓ Organic waste | ➡ | Anaerobic Digestion Plant |
| | ➡ | Composting |
| ✓ Construction &
Demolition Waste | ➡ | Recycling of C&D Waste |

Proposed Project Interventions

- ☐ **Engineered Landfill Site (LFS)**
 - ✓ Landfill Gas Recovery System
 - ✓ Leachate Management System
- ☐ **Remediation of Existing Dumpsite**
- ☐ **Remediation of Existing Dumpsite**
- ☐ **Carbon Credits**
- ☐ **Solarization**
- ☐ **BWMC office Commercial Building**

Project Benefits





PC-1

**DEVELOPING RESILIENT ENVIRONMENT AND ADVANCING
MUNICIPAL SERVICES (DREAMS) - I**

ORIGINAL APPROVED COST	PKR Million. 64,480.646/-
ORIGINAL APPROVED GESTATION	5 Year(s) (60 Months) Till December 2028
APPROVAL FORUM	ECNEC

1. NAME OF THE PROJECT

Developing Resilient Environment and Advancing Municipal Services (DREAMS-I)

2. LOCATION OF THE PROJECT

2.1. DISTRICT(S)

I. BAHAWALPUR, RAWALPINDI



To
Main PC-1 of DREAMS-I

**DEVELOPING RESILIENT ENVIRONMENT AND ADVANCING
MUNICIPAL SERVICES-I (DREAMS-I)**

**Project Component
for**

**INTEGRATED SOLID WASTE MANAGEMENT (ISWM) SYSTEM FOR
BAHAWALPUR**

Estimated Cost PKR. 11,562.743 million

**PROJECT DURATION 60 MONTHS
2024 - 2028**



Project Cost

Integrated Solid Waste Management System for
Bahawalpur

USD 40.35 million

Financials

Project Component	Total (PKR)
Primary and Secondary Waste Collection Machinery	2,095.243
Material Recovery System	2,389.089
Engineered Landfill	3,444.195
Other Components	1,981.404
O&M Cost for 3 years	1,378.750
Grand Total (Million PKR)	11,288.681
Grand Total (mil. USD)	40.35



Current Status

Contract Modality		Yellow Book under FIDIC conditions of contract
Duration	Design: 06 Months Build : 30 Months	
Scope	• Design of ISWM system • Procurement of Machinery and Equipment • Construction of ISWM Infrastructure • Capacity Building of BWMC Staff for operations	
Status	The Bidding Document under Yellow Book has been finalized and approved from ADB.	
Tender Bid Opening Date	15 January 2026	



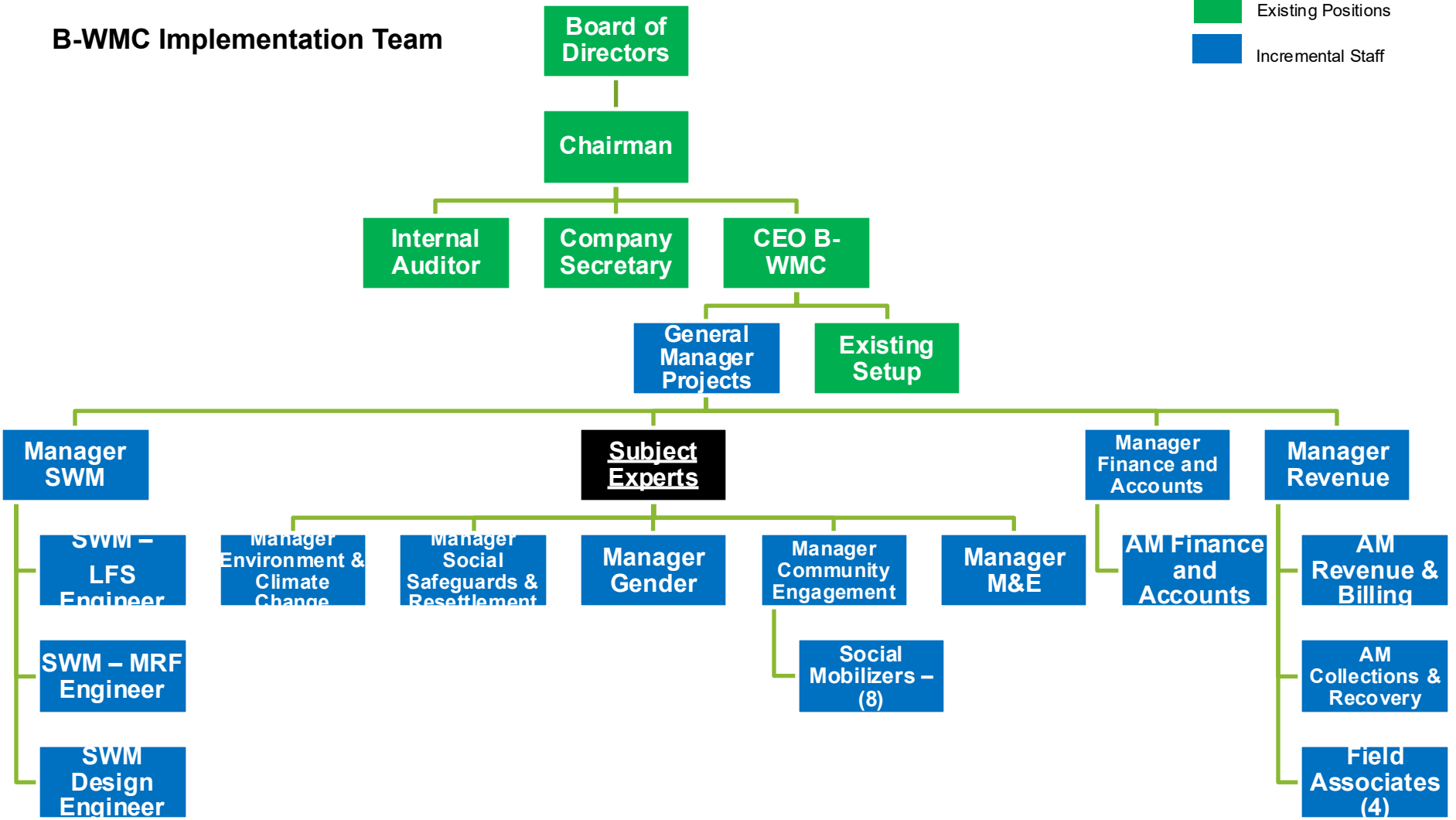
Scope of Bid Document (FIDIC Yellow Book)

Component	Details
Design, Procurement and Construction of ISWM System	• Primary & Secondary Collection Machinery & Equipment • Material Recovery Facility • Solid Waste Treatment Facilities (Anaerobic Digestion Plant, Compost Plant, Plastic Recycling Plant) • Engineered Landfill Site • Mitigation of Existing Dumpsite • Mixed Use Commercial Building including Parking for Machinery • Solarization of All Facilities • Carbon Credits
Capacity Building/Training of BWMC Staff (90 Days)	• Onsite Training for MRF, Landfill Operations for similar nature Infrastructure • Onsite Training of Staff for Waste Treatment Technologies • Training on Landfill Gas Collection and Leachate Treatment with Environment Compliance

BWMC - City Implementation Team (CIT)

B-WMC Implementation Team

 Existing Positions
 Incremental Staff



Strengthening BWMC



Current Capabilities



- ✓ **Daily Waste Collection:** appx 325 tons in Bahawalpur
- ✓ **Workforce:** Approximately 700 workers
- ✓ **Vehicles:** appx 110 operational vehicles

Enhancement Initiatives

Strategic support to build upon existing efforts, modernize systems, and systems, and expand solid waste management capabilities.



Service Delivery

- > Improve collection efficiency
- > Expand coverage areas



Policy Framework

- > Strengthen regulatory systems
- > Update operational guidelines



Institutional Capacity

- > Train management personnel
- > Upgrade administrative systems



System Modernization

- > Introduce modern equipment
- > Implement waste reduction programs

Public Awareness Campaigns

Improving Urban Community Practices

DREAMS Project implements public awareness and communication campaigns to improve behaviors and practices related to solid waste management among urban communities in Bahawalpur, focusing on education and engagement to foster responsible waste management habits.

Communication Strategies



Community Outreach

Direct engagement with residents to educate on proper waste segregation



Film Screenings

Utilizing film screenings as an effective medium to convey messages and stimulate discussion



Educational Workshops

Interactive sessions to teach proper waste management techniques



Expected Outcomes

- ✓ Improved public awareness about proper waste disposal
- ✓ Behavior change toward responsible waste management
- ✓ Enhanced community participation in waste management initiatives
- ✓ Improved environmental health outcomes in urban areas

Community Engagement Initiatives

Direct Community Engagement

DREAMS Project emphasizes direct community engagement and educational interventions to actively involve residents in solid waste management.



Community Education

Active involvement of residents in waste management processes through targeted educational programs



Awareness Campaigns

Public awareness initiatives to improve behaviors related to waste disposal and environmental cleanliness

Expected Outcomes

- ✓ Improved adoption of better waste disposal habits
- ✓ Enhanced community ownership of waste management
- ✓ Sustainable behavior change for environmental cleanliness



Film Screening Initiative

A powerful educational tool to drive behavior change through visual storytelling and community engagement.

Educational Content

Demonstrates proper waste segregation techniques and environmental impact

Community Discussion

Facilitates dialogue on local waste management challenges



Chand Bagh Film Screening

Special film screenings at Chand Bagh to maximize waste segregation initiative participation in Bahawalpur, with Q&A sessions and practical demonstrations



Promotion Strategy

Targeted outreach to maximize attendance and utilization of Chand Bagh screening facilities for waste segregation education in Bahawalpur

Expected Impact

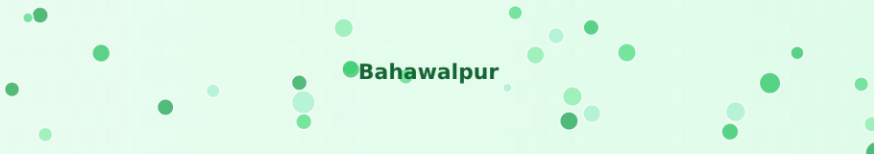
Transforming Urban Services in Pakistan

DREAMS Project partnership between ADB and UNICEF aims to create lasting positive impacts through sustainable and inclusive municipal services.

Target Population

1.5 million urban dwellers

Reaching communities in Bahawalpur



Improved Service Delivery

Enhanced quality, coverage, and reliability of water, sanitation, and waste management services



Social Inclusion

Particularly empowering children and women through improved access to essential services



Environmental Resilience

Reducing vulnerability to environmental degradation and climate change impacts

Long-term Benefits

- ✔ Creation of sustainable, low-carbon municipal infrastructure
- ✔ Strengthened institutional capacity for long-term service delivery
- ✔ Improved public health outcomes through better sanitation
- ✔ Enhanced quality of life in urban centers

Blue and Green Bin Initiative

Waste Segregation System

The Blue and Green Bin initiative is a cornerstone of the DREAMS Project's waste management strategy in Bahawalpur. This color-coded system makes waste segregation simple and intuitive for residents, promoting responsible waste disposal practices and enabling efficient recycling and composting processes.



Green Bin

Organic Waste

- Food scraps and leftovers
- Fruit and vegetable peels
- Garden waste and trimmings
- Coffee grounds and tea leaves



Composted into fertilizer



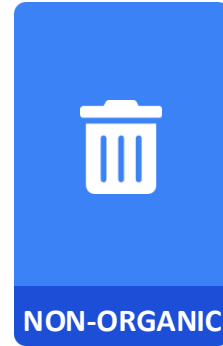
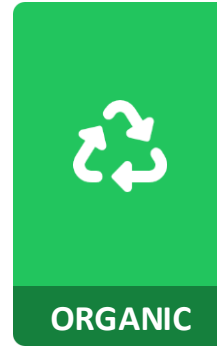
Blue Bin

Non-organic Waste

- Paper and cardboard
- Plastic containers and bottles
- Glass bottles and jars
- Metal cans and foil



Recycled or properly disposed



Environmental and Economic Benefits

- ✓ Reduces landfill waste by up to 50%
- ✓ Produces organic compost for agriculture
- ✓ Creates revenue from recyclable materials
- ✓ Reduces greenhouse gas emissions



Innovative Support Ideas

- Seed Pencils Initiative
Biodegradable writing tools with seed pockets
- Art from Waste
School competitions using recyclable materials



Mobile Waste Tracker
App with rewards for proper waste segregation



Clean Block Award
Monthly recognition for neighborhood participation

Training Methodology

Behavior Change Framework

DREAMS Project employs a comprehensive training methodology designed to drive sustainable behavior change in waste management practices. Our approach combines visual storytelling, practical demonstrations, and community participation to ensure lasting adoption of the Blue-Green Bin system throughout Bahawalpur.

Film Screening as a Cornerstone Strategy



Educational Film Series

Visual storytelling to demonstrate proper waste segregation

- ✓ Short, impactful (30 minutes & 16 minutes)
- ✓ Local context and characters for relatability
- ✓ Available in multiple local language & international
- ✓ Interactive elements for audience engagement



Screening Schedule

Monthly schedules in schools, bi-weekly in community centers, and weekend screenings in public spaces with promotional incentives



Training Methodology

Training Implementation Process

- 1 Community Film Screenings**
Public screenings in community centers followed by Q&A sessions
- 2 Hands-on Workshops**
Practical demonstrations using actual blue and green bins in neighborhoods
- 3 Train-the-Trainer Programs**
Building local capacity by training community champions to continue education
- 4 School Education Programs**
Engaging students as agents of change to influence household practices
- 5 Follow-up and Monitoring**
Regular community visits to assess adoption and address challenges



Key Accomplishments

Seed Pencil Initiative

After using the pencil, the biodegradable casing containing seeds can be planted to grow herbs or flowers, reinforcing the waste-to-resource concept



Organics



Paper & Cardboard

Target Audiences & Engagement Incentives



Households

stickers (**Green-Organic** & **Blue-Inorganic**)



Businesses

Recognition certificates & green business badges, pen



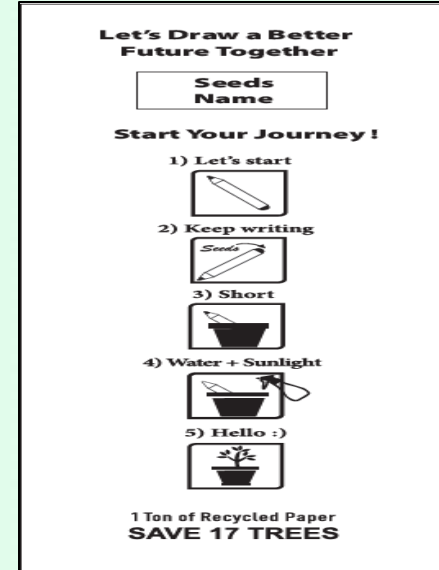
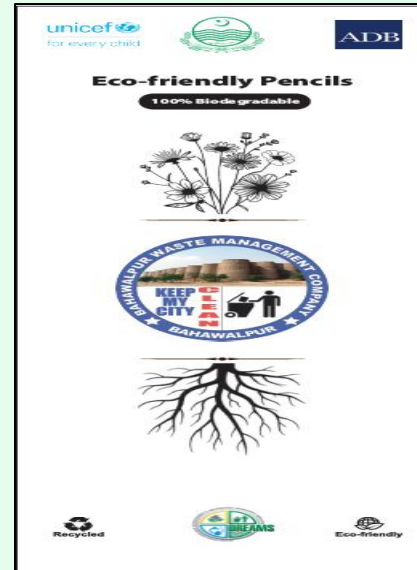
Schools

Biodegradable pencils with seed pockets for students



Institutions

Waste reduction challenge with prizes



Becoming Waste Warriors: Student Engagement

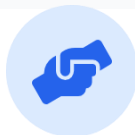


Every student can become a **"Waste Warrior & Ambassador"** and help transform Bahawalpur into a cleaner, healthier city.



Educate & Inform

- ✓ Integrate waste segregation into curriculum
- ✓ Organize educational workshops and assemblies
- ✓ Create visual learning materials for classrooms



Engage & Participate

- ✓ Join the BWMC Waste Warrior program
- ✓ Participate in school-wide clean-up drives
- ✓ Help maintain the cleanliness of color-coded bins



Lead & Inspire

- ✓ Start a student environmental club
- ✓ Organize awareness campaigns
- ✓ Share knowledge with friends and family



Student Roles as Waste Warriors & Ambassador



Green Team: Organic Expert



Blue Team: Paper Patrol



Yellow Team: Plastic Police



Red Team: Hazard Hunter

The Waste Warrior & Ambassador Pledge

I promise to:



Always use the correct bin



Keep bins clean and tidy



Encourage others to participate

Join us in making Bahawalpur a cleaner, healthier city!



Q&A session follows

UNICEF - ADB, Smart Waste Management Initiative

BWMC | Smart Waste Management





THANK YOU!